



An AEP Company

BOUNDLESS ENERGY™

Indiana Michigan Power
Cook Nuclear Plant
One Cook Place
Bridgman, MI 49106
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March 27, 2025

AEP-NRC-2025-10
10 CFR 50.73

Docket No.: 50-316

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
11555 Rockville Pike
Rockville, MD 20852

Donald C. Cook Nuclear Plant Unit 2
LICENSEE EVENT REPORT 316/2024-004-01

2AB Emergency Diesel Generator Inoperable for longer than allowed by Technical Specifications

In accordance with 10 CFR 50.73, Licensee Event Report (LER) System, Indiana Michigan Power Company, the licensee for Donald C. Cook Nuclear Plant Unit 2, is submitting as an enclosure to this letter the following report:

LER 316/2024-004-01: 2AB Emergency Diesel Generator Inoperable for longer than allowed by Technical Specifications

This LER is a supplement to LER 316/2024-004-00 (ML24316A006), which was previously submitted by letter dated November 11, 2024. The enclosed LER has been revised to update the Apparent and Root causes of the event and the associated corrective actions.

There are no commitments contained in this submittal.

Should you have any questions, please contact Mr. Michael K. Scarpello, Regulatory Affairs Director, at (269) 466-2649.

Sincerely,

A handwritten signature in blue ink, reading "Scott A. Dailey".

Scott A. Dailey
Site Vice President

MPH/sjh

Enclosure: Licensee Event Report 316/2024-004-01: 2AB Emergency Diesel Generator Inoperable for longer than allowed by Technical Specifications

c: EGLE - RMD/RPS
J. B. Giessner -- NRC Region III
NRC Resident Inspector
N. Quilico -- MPSC
R. M. Sistevaris -- AEP Ft. Wayne
S. P. Wall, NRC Washington D.C.
A. J. Williamson -- AEP Ft. Wayne

Enclosure to AEP-NRC-2025-10

Licensee Event Report 316/2024-004-01: 2AB Emergency Diesel Generator Inoperable for
longer than allowed by Technical Specifications

NRC FORM 366 (04-02-2024)		U.S. NUCLEAR REGULATORY COMMISSION		APPROVED BY OMB: NO. 3150-0104		EXPIRES: 04/30/2027					
LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/) Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.											
1. Facility Name Donald C. Cook Nuclear Plant Unit 2				☒ 050 ☐ 052		2. Docket Number 00316					
						3. Page 1 OF 3					
4. Title 2AB Emergency Diesel Generator Inoperable for longer than allowed by Technical Specifications											
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☒ 050	Docket Number
09	12	2024	2024	- 004 -	01	03	27	2025	Cook Nuclear Plant Unit 1		315
									Facility Name	☐ 052	Docket Number
9. Operating Mode 1						10. Power Level 100					
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)											
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.1200(a)	
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		☐ 73.1200(b)	
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		☐ 73.1200(c)	
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.36(c)(2)		☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		☐ 73.1200(d)	
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.46(a)(3)(ii)		☒ 50.73(a)(2)(v)(A)		10 CFR Part 73		☐ 73.1200(e)	
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.69(g)		☒ 50.73(a)(2)(v)(B)		☐ 73.77(a)(1)		☐ 73.1200(f)	
☐ 20.2203(a)(2)(iii)				☐ 50.73(a)(2)(i)(A)		☒ 50.73(a)(2)(v)(C)		☐ 73.77(a)(2)(i)		☐ 73.1200(g)	
☐ 20.2203(a)(2)(iv)				☒ 50.73(a)(2)(i)(B)		☒ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)		☐ 73.1200(h)	
☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)					
☐ OTHER (Specify here, in abstract, or NRC 366A).											
12. Licensee Contact for this LER											
Licensee Contact Michael K. Scarpello, Regulatory Affairs Director									Phone Number (Include area code) (269) 466-2649		
13. Complete One Line for each Component Failure Described in this Report											
Cause	System	Component	Manufacturer	Reportable to IRIS		Cause	System	Component	Manufacturer	Reportable to IRIS	
X	DG	DRU	WW	Y							
14. Supplemental Report Expected						15. Expected Submission Date			Month	Day	Year
☒ No		☐ Yes (If yes, complete 15. Expected Submission Date)									
16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)											
On May 21, 2024, with Cook Nuclear Plant (CNP) Units 1 and 2 in Mode 1 at 100% power, the 2AB EDG failed to reach the required frequency during the slow speed start surveillance, per the required acceptance criteria range of 59.5 to 60.4 Hz. The failure to reach the required frequency was originally attributed to corroded connections on the Minimum Speed Threshold and Slow Start control relays. On July 23, 2024, the 2AB EDG, once again, failed to reach the required frequency during a surveillance run. The cause of the failure was determined to be an intermittent failure of the Digital Reference Unit (DRU). Due to the similar symptoms between the two failures of the 2AB EDG, it was determined that the 2AB EDG failure on May 21, 2024, was also due to a failure of the DRU. Due to the discovery that both EDG failures were a result of a failure of the DRU, the 2AB EDG was determined to be inoperable from the time the original condition was discovered on 05/21/2024 at 2115, until the time it successfully passed a surveillance after the DRU was replaced on 7/24/2024 at 2300. Therefore, the identified condition is reportable, for both units, in accordance with 10 CFR 50.73(a)(2)(i)(B) as a Condition prohibited by Technical Specifications. Furthermore, during this period of time, there were instances on Unit 2 where concurrent with the 2AB EDG, the 2CD EDG was inoperable, which results in being reportable per 10 CFR 50.73(a)(2)(v) as an Event or Condition which could have prevented the fulfillment of a Safety Function. The Root Cause for the failure to properly identify and correct the May 21, 2024 EDG surveillance failure was a lack of proper oversight and proper use of knowledge worker human performance tools during the Failure Investigation Process (FIP). Corrective Actions include formalization and improvements in qualifying FIP team management sponsors and team leaders.											

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME

Donald C. Cook Nuclear Plant Unit 2



050



052

2. DOCKET NUMBER

00316

3. LER NUMBER

YEAR

2024

SEQUENTIAL
NUMBER

004

REV
NO.

01

NARRATIVE**EVENT DESCRIPTION**

On May 21, 2024, with Cook Nuclear Plant (CNP) Units 1 and 2 in Mode 1 at 100% power, the 2AB Emergency Diesel Generator (EDG) [DG] failed to reach the required frequency during the slow speed start surveillance, per the required acceptance criteria range of 59.5 to 60.4 Hz. The failure to reach the required frequency was originally attributed to corroded connections on the Minimum Speed Threshold and Slow Start control relays.

On July 23, 2024, the 2AB EDG, once again, failed to reach the required frequency during a surveillance run. The cause of the failure was determined to be an intermittent failure of the Digital Reference Unit (DRU). Due to the similar symptoms between the two failures of the 2AB EDG, it was determined that the 2AB EDG failure on May 21, 2024, was also due to the failure of the DRU.

Unit 2 Limiting Condition of Operation (LCO) 3.8.1 requires two diesel generators (DGs) capable of supplying the onsite Class 1E power distribution subsystem(s) to be OPERABLE in MODE 1. Additionally, Unit 1 LCO 3.8.1 requires the Unit 2 EDG(s) to be capable of supporting the associated equipment required to be OPERABLE by LCO 3.7.8, Essential Service Water (ESW) System [BI]. If both Unit 1 ESW trains are cross-tied to Unit 2, then the two Unit 2 EDGs are required to be OPERABLE. The maximum COMPLETION TIME allowed to restore an inoperable EDG to OPERABLE status in MODE 1 is 14 days. With this COMPLETION TIME not met, then the affected Unit is required to be in MODE 3 in 6 hours and MODE 5 in 36 hours.

Due to the discovery that both EDG failures were a result of a failure of the DRU, the 2AB EDG was determined to be inoperable from the time the original condition was discovered on 05/21/2024 at 2115, until the time it successfully passed a surveillance after the DRU was replaced on 7/24/2024 at 2300. The period of time that the 2AB EDG was determined to be INOPERABLE exceeds the time allowed by LCO 3.8.1. Therefore, the identified condition is reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) as a Condition prohibited by Technical Specifications, for Units 1 and 2.

Additionally, during this period of time, it was determined that the redundant EDG (2CD) was INOPERABLE for surveillance and maintenance activities during the following times:

05/22/2024 21:08 - 05/22/2024 22:33 U2 CD INOPERABLE for U1 ESW surveillance

05/30/2024 07:05 - 05/30/2024 14:50 U2 CD INOPERABLE for maintenance

06/03/2024 20:15 - 06/04/2024 01:25 U2 CD INOPERABLE for surveillance activities

07/09/2024 19:34 - 07/10/2024 00:15 U2 CD INOPERABLE for surveillance activities

07/10/2024 20:52 - 07/10/2024 21:46 U2 CD INOPERABLE for U2 ESW surveillance

Additionally, LCO 3.8.1 contains a Prescribed Action to declare supported systems inoperable upon discovery of the inoperability of redundant required features concurrent with DG inoperability, with a Completion Time of 4 hours. During the time that the 2AB EDG was determined to be inoperable, there were instances where supported systems should have been declared inoperable. Therefore, during these instances, as well as the 5 instances that 2CD EDG was INOPERABLE for surveillance and maintenance activities, the identified condition is reportable per 10 CFR 50.73(a)(2)(v) as an Event or Condition which could have prevented the fulfillment of a Safety Function for Unit 2.

COMPONENT

2-DGAB-DRU - 2AB EDG DRU

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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NARRATIVE**CAUSE OF THE EVENT**

An Apparant Cause Evaluation (ACE) was performed to determine the cause of the 2AB EDG failures, and determined that Engineers and Technical Staff did not ensure assumptions and judgement were fully documented and conservative, clearly communicated to decision makers, or validated when investigating and evaluating irreproducible failure modes on the EDG Digital Reference Units (DRUs).

Additionally, a Root Cause Evaluation (RCE) was performed to determine why the station failed to properly identify the correct cause of the May 2024 2AB Emergency Diesel Generator failure, and determined that the May 2024 Failure Investigation Process (FIP) lacked proper oversight to ensure the appropriate and proper use of knowledge worker human performance tools.

CORRECTIVE ACTIONS

Corrective Actions to Prevent Recurrence (CAPR) include:

Develop qualifications for Duty Station Managers to serve as FIP Management Sponsors and FIP Designated Challengers, and for personnel to serve as FIP Team Leaders.

Add a reference in the FIP Procedures to use PMP-4010-HUT-002, Human Performance Tools for Knowledge Workers, to validate assumptions.

ASSESSMENT OF SAFETY CONSEQUENCES**NUCLEAR SAFETY**

A probabilistic risk assessment (PRA) was performed to estimate the impact of the 2AB EDG inoperability on Core Damage Frequency (CDF) and Large Early Release Frequency (LERF). Further review of the condition was performed to identify appropriate PRA mapping for this assessment. This review concluded that for the frequencies observed during the failed surveillances, the 2AB EDG would have been able to meet its PRA success criteria. However, this review was unable to eliminate the potential for the failed DRU to result in an EDG failure during its mission time. To capture these insights, this assessment was performed by assuming a higher failure probability for the 2AB EDG.

The evaluation used the average test/maintenance internal events and fire PRAs to estimate an increase in CDF and LERF over an assumed 63-day exposure time. These results are within the bounds for "Very Low Safety Significance" as described within NRC Inspection Manual Chapter 0609.

INDUSTRIAL SAFETY

There was no actual or potential industrial safety hazard resulting from the 2AB EDG Inoperability.

RADIOLOGICAL SAFETY

There was no actual or potential radiological safety hazard resulting from the 2AB EDG Inoperability.

PREVIOUS SIMILAR EVENTS

A review of Licensee Event Reports for the past five years found no similar events.